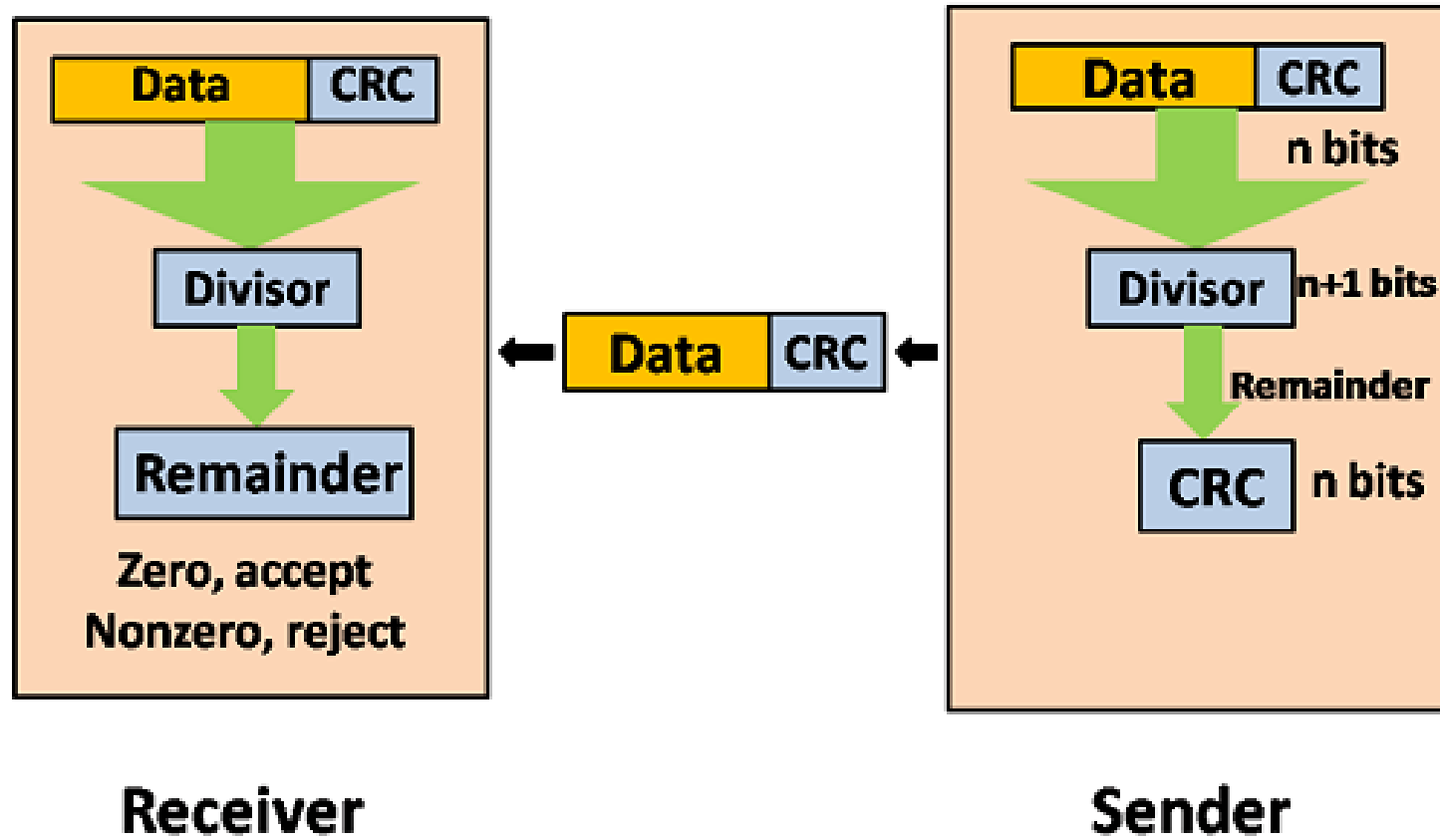
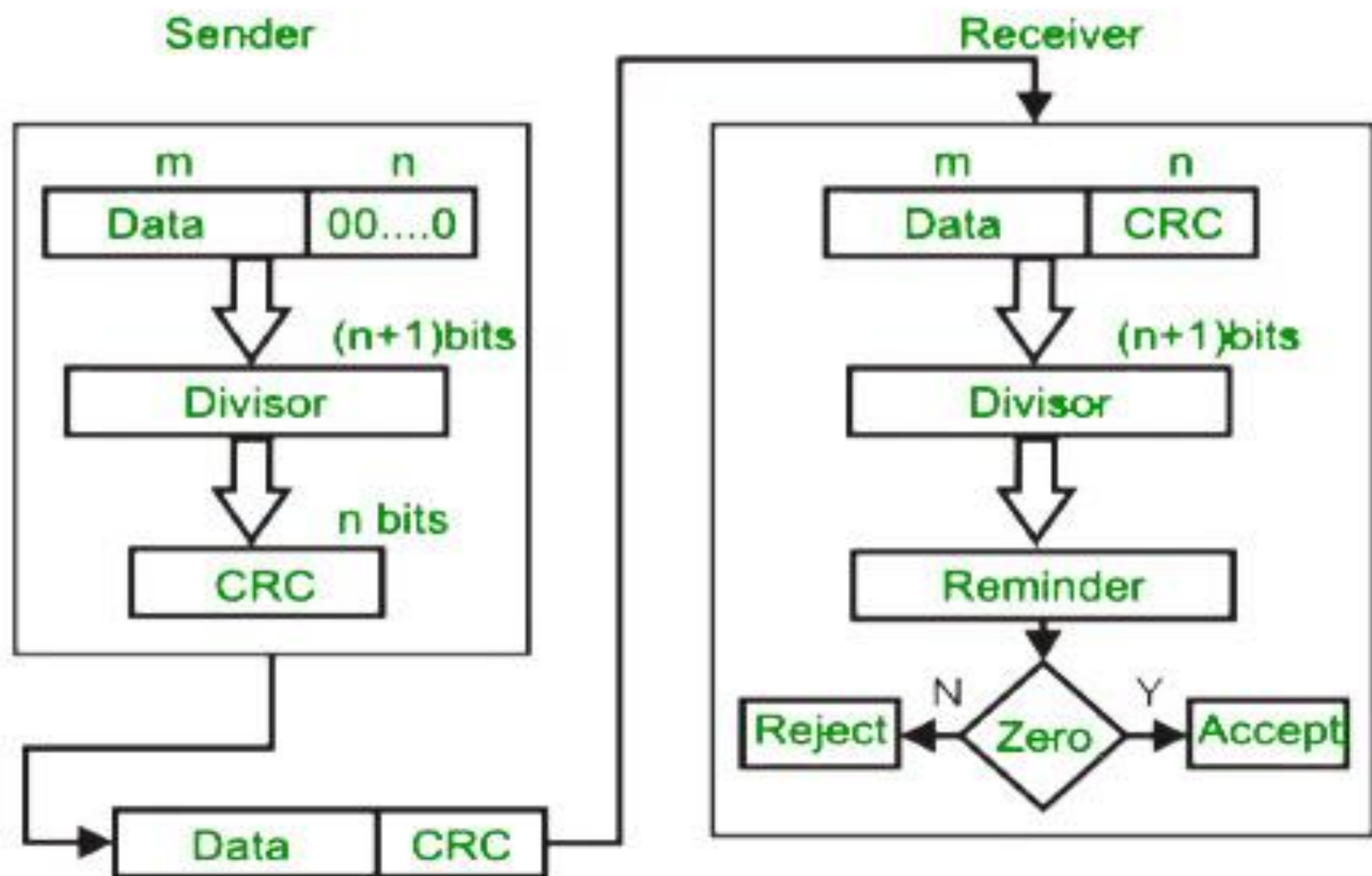


CRC

ANUSHA N





CRC

- **Sender Side (CRC Generator and Modulo Division):**

1. The first step is to add the no. of zeroes to the data to be sent, calculated using $k-1$ (k - is the bits obtained through the polynomial equation.)
2. Applying the Modulo Binary Division to the data bit by applying the XOR and obtaining the remainder from the division
3. The last step is to append the remainder to the end of the data bit and share it with the receiver.

- **Receiver Side (Checking for errors in the received data):**

- To check the error, perform the Modulo division again and check whether the remainder is 0 or not,
 1. If the remainder is 0, the data bit received is correct, without any errors.
 2. If the remainder is not 0, the data received is corrupted during transmission.

CRC

- Step-01: Calculation Of CRC At Sender Side-
- At sender side,
- A string of n 0's is appended to the data unit to be transmitted.
- Here, n is one less than the number of bits in CRC generator.
- Binary division is performed of the resultant string with the CRC generator.
- After division, the remainder so obtained is called as **CRC**.
- It may be noted that CRC also consists of n bits.

CRC

Step-02: Appending CRC To Data Unit-

At sender side,

The CRC is obtained after the binary division.

The string of n 0's appended to the data unit earlier is replaced by the CRC remainder.

Step-03: Transmission To Receiver-

The newly formed code word (Original data + CRC) is transmitted to the receiver.

Step-04: Checking at Receiver Side-

At receiver side,

The transmitted code word is received.

The received code word is divided with the same CRC generator.

On division, the remainder so obtained is checked.

Receiver assumes that some error occurred in the data during the transmission.

Receiver rejects the data and asks the sender for retransmission.

CRC

The following two cases are possible-

Case-01: Remainder = 0

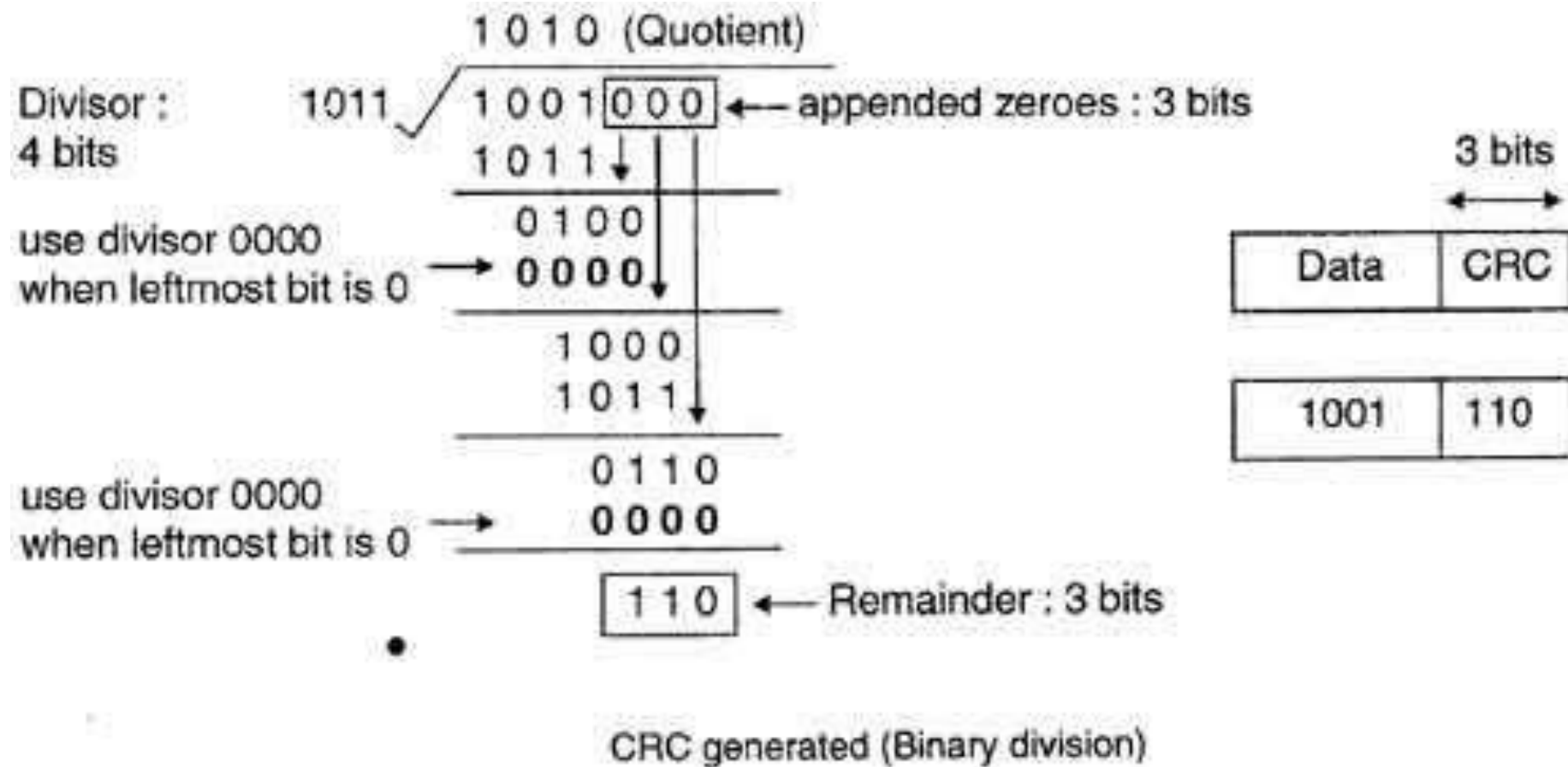
If the remainder is zero,
Receiver assumes that no error occurred in the data during the transmission.
Receiver accepts the data.

Case-02: Remainder $\neq 0$

If the remainder is non-zero,
Receiver assumes that some error occurred in the data during the transmission.
Receiver rejects the data and asks the sender for retransmission.

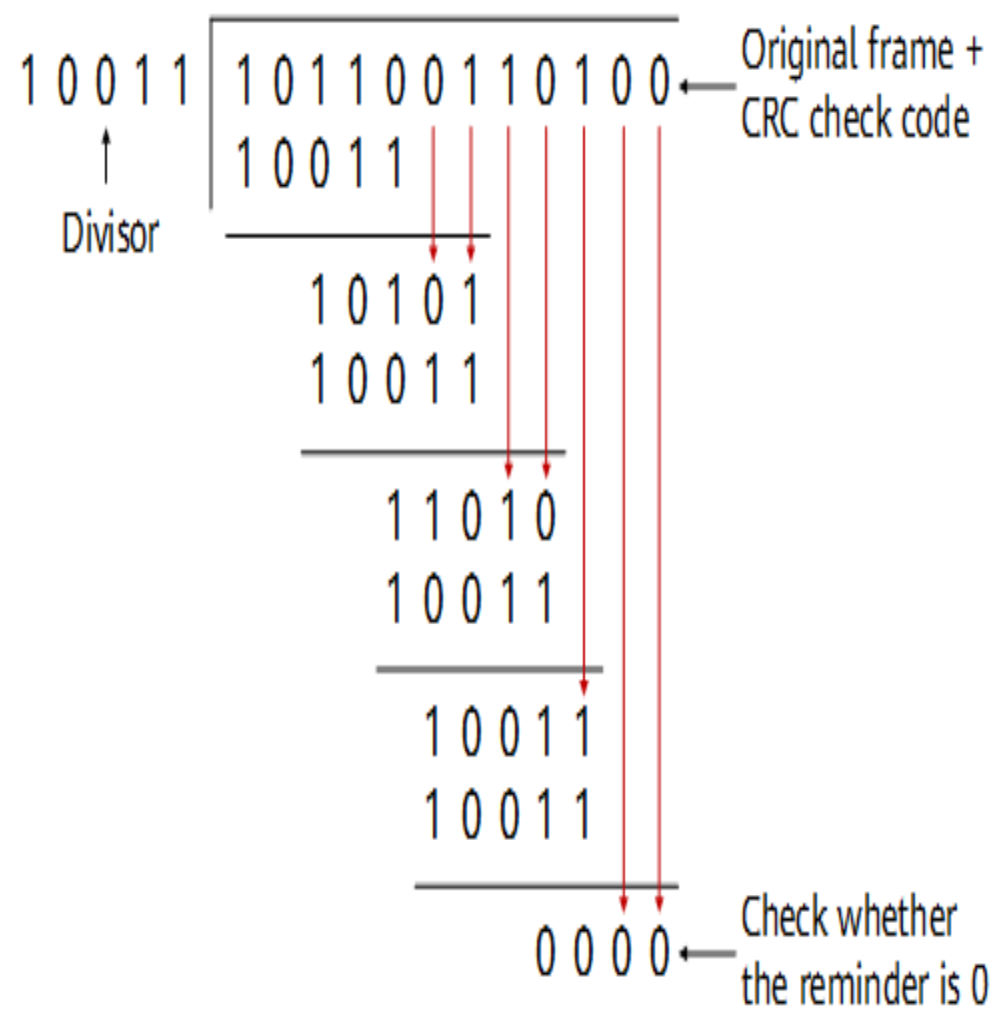
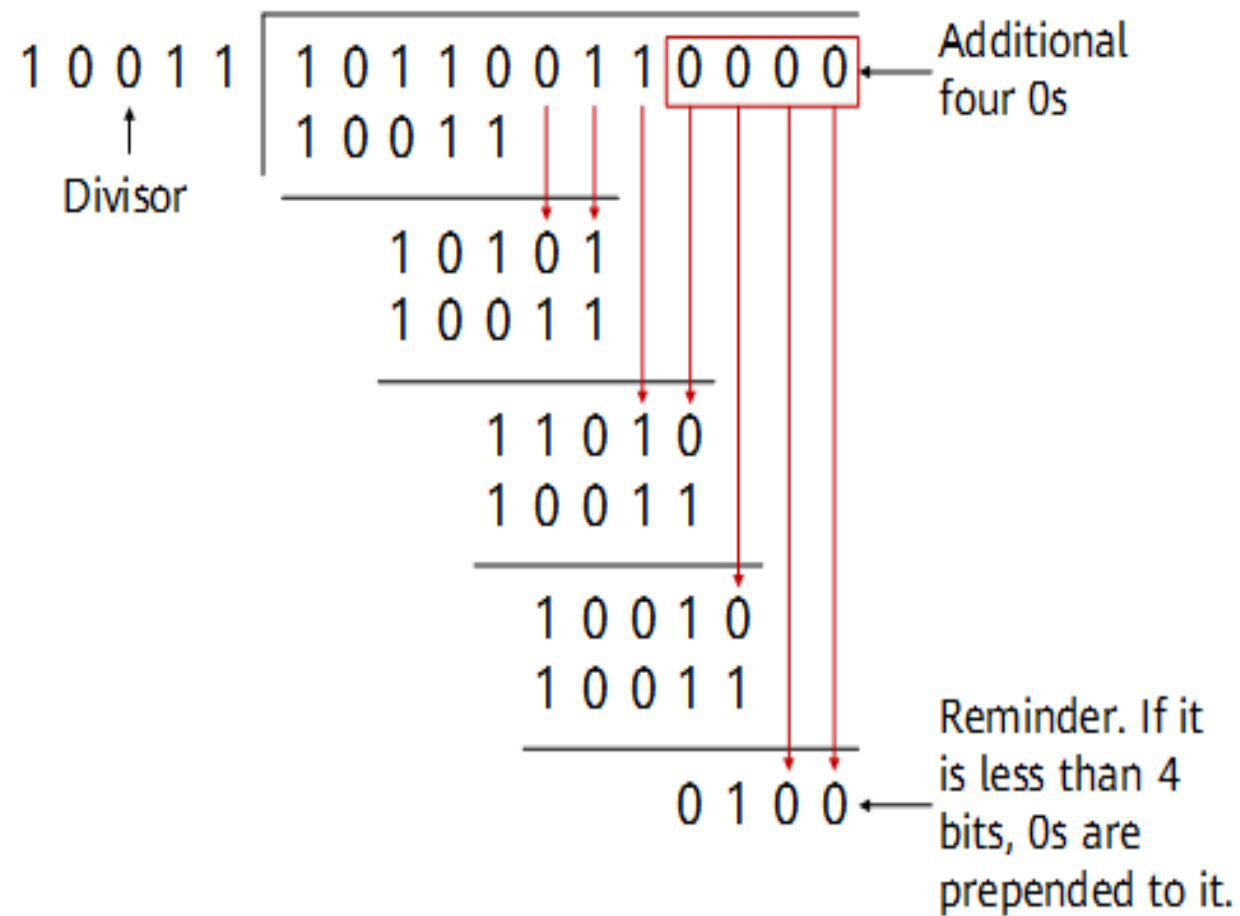
-

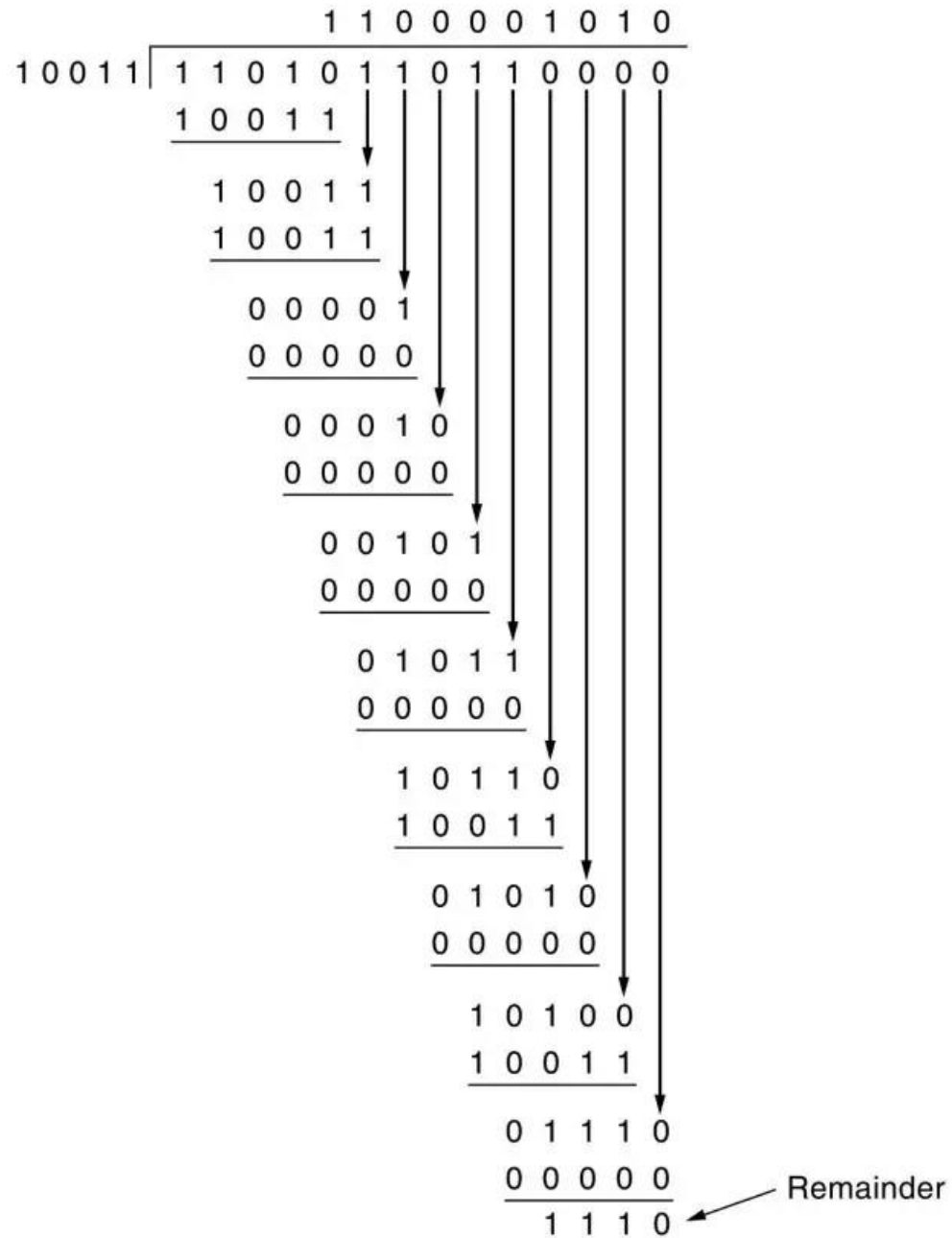
PRACTICE PROBLEM 1



Practice Problems

1. Data -- 1101011011 ; Divisor-- 10011
2. Data --10011101 ; Divisor-- 1001
3. Data-- 100100 Divisor-1101
4. Data-1010000;Divisor -1001





From here, CRC = 1110.

Now,

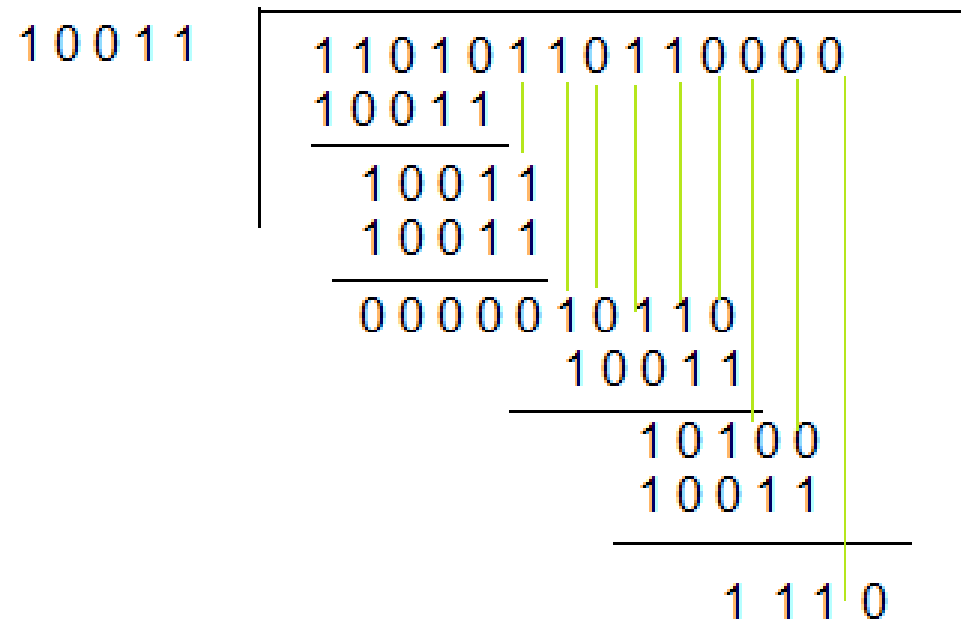
- The code word to be transmitted is obtained by replacing the last 4 zeroes of 1101011011**0000** with the CRC.
- Thus, the code word transmitted to the receiver = 1101011011**1110**.

	111101	----->	Quotient
Divisor	← 1101		
	100100000	----->	Dividend
	<u>1101</u>		
	1000		
	<u>1101</u>		
	1010		
	<u>1101</u>		
	1110		
	<u>1101</u>		
	0110		
	<u>0000</u>		
	1100		
	<u>1101</u>		
	001	----->	Remainder

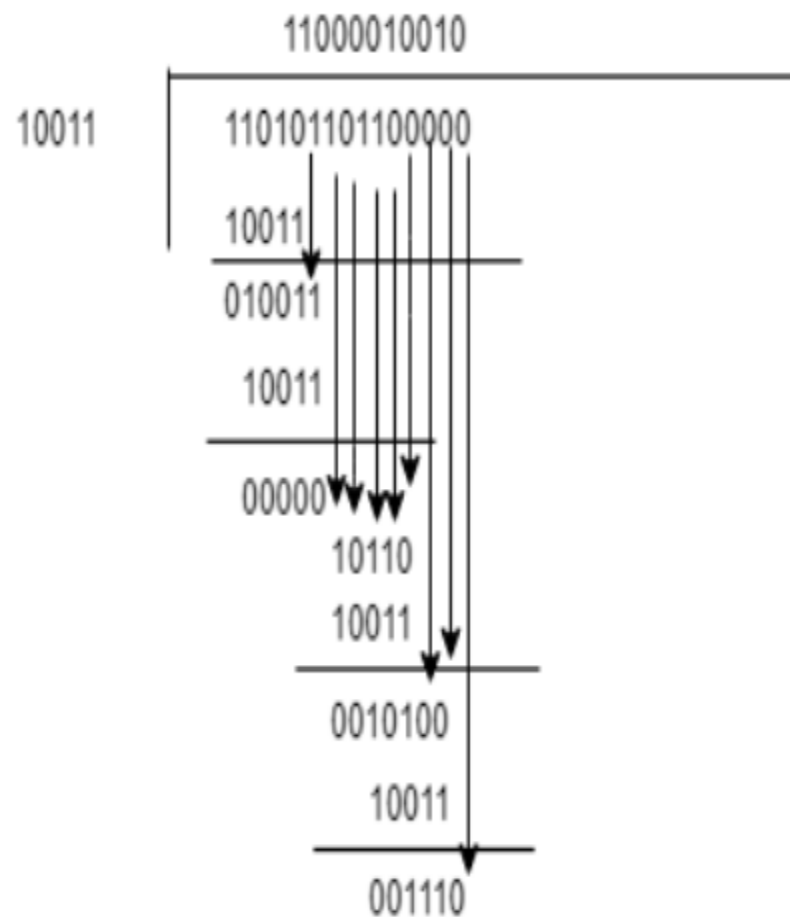
	111101	----->	Quotient
Divisor	← 1101		
	100100001	----->	Dividend
	<u>1101</u>		
	1000		
	<u>1101</u>		
	1010		
	<u>1101</u>		
	1110		
	<u>1101</u>		
	0110		
	<u>0000</u>		
	1101		
	<u>1101</u>		
	000	----->	Remainder

In CRC checksum method, assume that given frame for transmission is 1101011011 and the divisor is 10011. After implementing CRC encoder, the encoded word sent from sender side is

In CRC checksum method, assume that given frame for transmission is 1101011011 and the divisor is 10011. After implementing CRC encoder, the encoded word sent from sender side is _



Explanation: Given frame for transmission is 1101011011 and generator polynomial is $G(x) = x^4 + x + 1$ i.e. 10011. We have to append 4 0's (1 0's less than divisor according to CRC):



We have to append 1110 to the frame. Now our frame for transmission is 11010110111110.

original message
1 0 1 0 0 0 0

@ means X-OR

Generator polynomial
 x^3+1
 $1.x^3+0.x^2+0.x^1+1.x^0$
CRC generator
1 0 0 1 4-bit

If CRC generator is of n bit then append $(n-1)$ zeros in the end of original message

Sender

1 0 0 1 | 1 0 1 0 0 0 0 0 0 0
@ 1 0 0 1
0 0 1 1 0 0 0 0 0 0
@ 1 0 0 1
0 1 0 1 0 0 0 0
@ 1 0 0 1
0 0 1 1 0 0 0
@ 1 0 0 1
0 1 0 1 0
@ 1 0 0 1
0 0 1 1

Message to be transmitted

1 0 1 0 0 0 0 0 0 0
+ 0 1 1
1 0 1 0 0 0 0 0 1 1

1 0 0 1 | 1 0 1 0 0 0 0 0 1 1
@ 1 0 0 1
0 0 1 1 0 0 0 0 1 1
@ 1 0 0 1
0 1 0 1 0 0 1 1
@ 1 0 0 1
0 0 1 1 0 1 1
@ 1 0 0 1
0 1 0 0 1
@ 1 0 0 1
0 0 0 0

Receiver

Zero means data is accepted

CRC USING POLYNOMIALS

ANUSHA N

CRC USING POLYNOMIALS

- Consider the CRC generator is $x^7 + x^6 + x^4 + x^3 + x + 1$.

$$\begin{array}{cccccccc} 1x^7 & + & 1x^6 & + & 0x^5 & + & 1x^4 & + & 1x^3 & + & 0x^2 & + & 1x^1 & + & 1x^0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 1 & & 1 & & 0 & & 1 & & 1 & & 0 & & 1 & & 1 \end{array}$$

- Thus, for the given CRC generator, the corresponding binary pattern is 11011011.
-

CRC USING POLYNOMIALS

- A pattern of 0s and 1s can be represented as a polynomial with coefficient of 0 and 1.
- Here, the power of each term shows the position of the bit and the coefficient shows the values of the bit.
- For example, if binary pattern is 100101,
- its corresponding polynomial representation is $x^5 + x^2 + 1$

Polynomial codes is that it produces short codes.

For example here a 6-bit pattern is replaced by 3 terms.

- In polynomial codes, the degree is 1 less than the number of bits in the binary pattern.
- The degree of polynomial is the highest power in polynomial. For example as shown in fig degree of polynomial $x^5 + x^2 + 1$ are 5. The bit pattern in this case is 6.
- Addition of two polynomials is based on modulo-2 method. In such as case, addition and subtraction is same.
- Addition or subtraction is done by combining terms and deleting pairs of identical terms. For example adding $x^5 + x^4 + x^2$ and $x^6 + x^4 + x^2$ give $x^6 + x^5$. The terms x^4 and x^2 are deleted.
- If three polynomials are to be added and if we get a same term three times, a pair of them is detected and the third term is kept. For example, if there is x^2 three times then we keep only one x^2

CRC USING POLYNOMIALS

- In case of multiplication of two polynomials, their powers are added. For example, multiplying $x^5 + x^3 + x^2 + x$ with $x^2 + x + 1$ yields:
- $(x^5 + x^3 + x^2 + x)(x^2 + x + 1)$
- $= x^7 + x^6 + x^5 + x^5 + x^4 + x^3 + x^4 + x^3 + x^2 + x^3 + x^2 + x$
- $= x^7 + x^6 + x^3 + x$
- In this, first polynomial is multiplied by all terms of second. The result is then simplified and pairs of equal terms are deleted.
- • In case of division, the two polynomials are divided as per the rules of binary division, until the degree of dividend is less than that of divisor.

CRC USING POLYNOMIALS

- The polynomial generator should have following properties:
 1. It should have at least two terms.
 2. The coefficient of the term x^0 should be 1.
 3. It should not be divisible by x .
 4. It should be divisible by $x+1$.

CRC USING POLYNOMIALS

Important Notes-

- If the CRC generator is chosen according to the above rules, then-
- CRC can detect all single-bit errors
- CRC can detect all double-bit errors provided the divisor contains at least three logic 1's.
- CRC can detect any odd number of errors provided the divisor is a factor of $x+1$.
- CRC can detect all burst error of length less than the degree of the polynomial.
- CRC can detect most of the larger burst errors with a high probability.

CRC USING POLYNOMIALS

- A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1 .

1.What is the actual bit string transmitted?

2.Suppose the third bit from the left is inverted during transmission. How will receiver detect this error?

- **Solution-**

-

- **Part-01:**

-

- The generator polynomial $G(x) = x^3 + 1$ is encoded as 1001.
- Clearly, the generator polynomial consists of 4 bits.
- So, a string of 3 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 10011101**000**.

$$\begin{array}{r}
 1001 \overline{) 10001100} \\
 \underline{100111101000} \\
 1001 \\
 \underline{00001} \\
 0000 \\
 \underline{0000} \\
 00011 \\
 0000 \\
 \underline{0000} \\
 00110 \\
 0000 \\
 \underline{0000} \\
 01101 \\
 1001 \\
 \underline{01000} \\
 1001 \\
 \underline{00010} \\
 0000 \\
 \underline{0000} \\
 00100 \\
 0000 \\
 \underline{0000} \\
 0100 \leftarrow \text{CRC}
 \end{array}$$

From here, CRC = 100.

Now,

- The code word to be transmitted is obtained by replacing the last 3 zeroes of 10011101**000** with the CRC.
- Thus, the code word transmitted to the receiver = 10011101**100**.

CRC USING POLYNOMIALS

- Part-02:
- According to the question,
- Third bit from the left gets inverted during transmission.
- So, the bit stream received by the receiver = 10111101100.
- Now,
- Receiver receives the bit stream = 10111101100.
- Receiver performs the binary division with the same generator polynomial as-

Part-02:

According to the question,

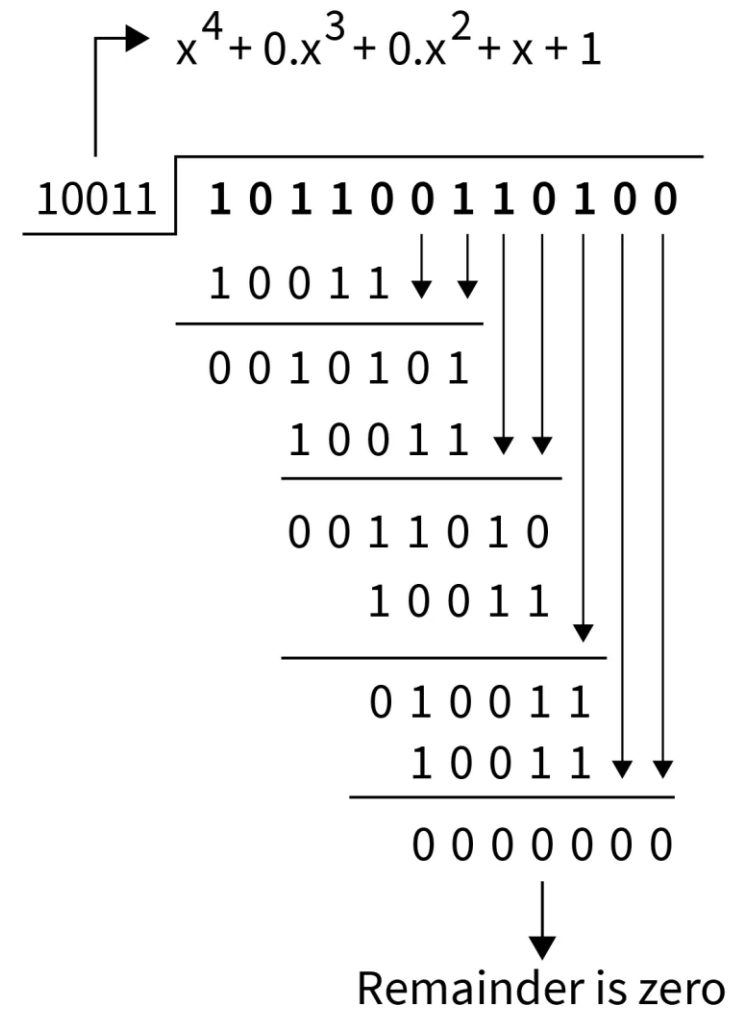
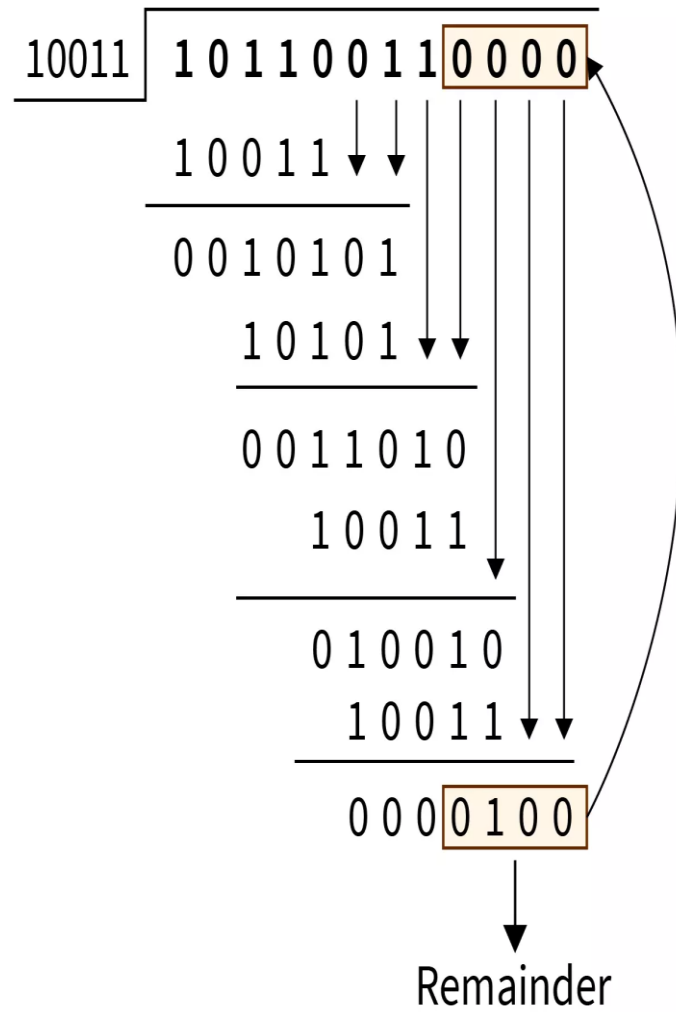
- Third bit from the left gets inverted during transmission.
- So, the bit stream received by the receiver = 10111101100.

Now,

- Receiver receives the bit stream = 10111101100.
 - Receiver performs the binary division with the same generator polynomial as-
- From here,
- The remainder obtained on division is a non-zero value.
 - This indicates to the receiver that an error occurred in the data during the transmission.
 - Therefore, receiver rejects the data and asks the sender for retransmission.

$$\begin{array}{r} 1001 \overline{) 10101000} \\ 1001 \\ \hline 00101 \\ 0000 \\ \hline 01011 \\ 1001 \\ \hline 00100 \\ 0000 \\ \hline 01001 \\ 1001 \\ \hline 00001 \\ 0000 \\ \hline 00010 \\ 0000 \\ \hline 00100 \\ 0000 \\ \hline 0100 \end{array} \leftarrow \text{Remainder}$$

- Given that the data stream is 10110011 and the generator polynomial is x^4+x+1



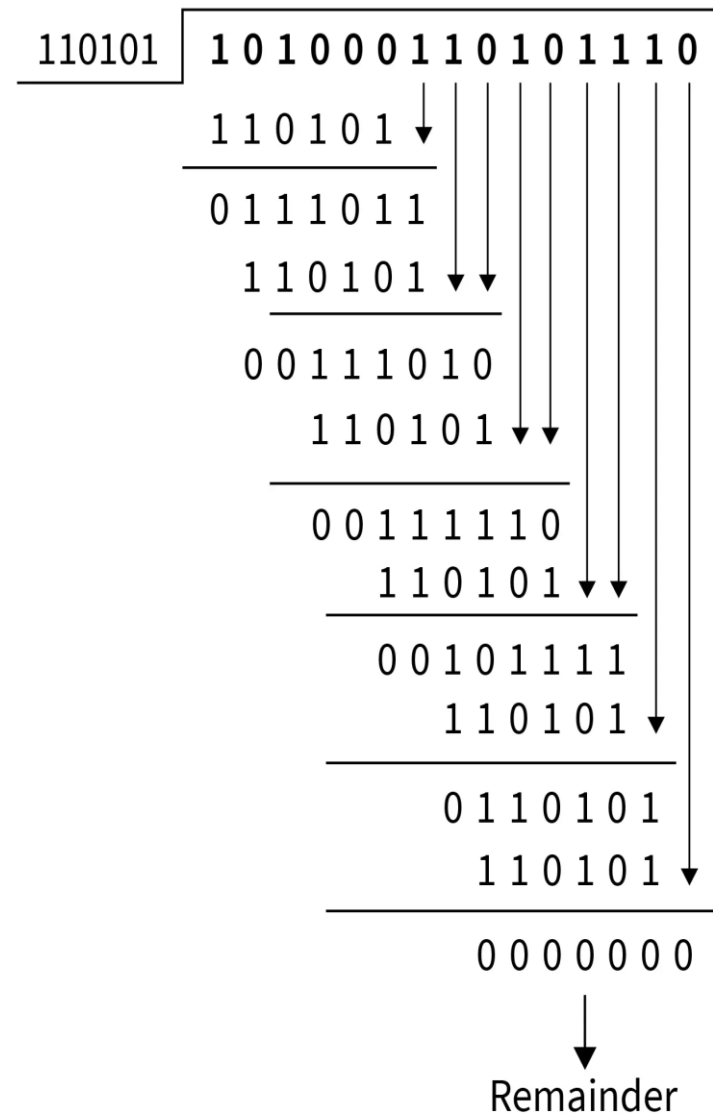
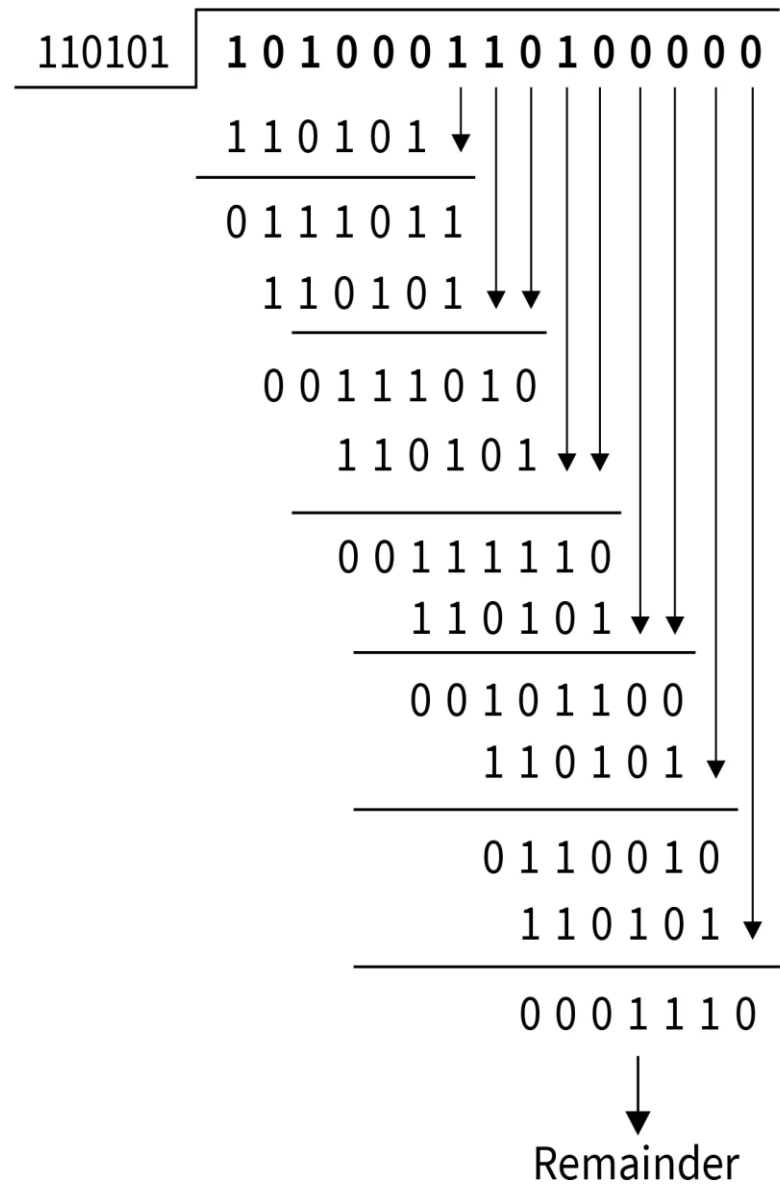
- A bit stream 1101011011 is transmitted using the standard CRC method. The generator polynomial is x^4+x+1 . What is the actual bit string transmitted?

- **Solution-**

-

- The generator polynomial $G(x) = x^4 + x + 1$ is encoded as 10011.
- Clearly, the generator polynomial consists of 5 bits.
- So, a string of 4 zeroes is appended to the bit stream to be transmitted.
- The resulting bit stream is 1101011011**0000**.

- Consider the message sender wants to send is 1010001101, and the generator polynomial is $x^5+x^4+x^2+1$. Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.



Suppose we want to transmit the message 11001001 and protect it from errors using the CRC polynomial x^3+1 . Use polynomial long division to determine the message that should be transmitted. Corrupt the left-most third bit of the transmitted message and show that the error is detected by the receiver using CRC technique.

- If we consider the data unit 1001 and divisor or polynomial generator 1011 their polynomial representation is:

Data 1001 $x^3 + 1$
 Division 1011 $x^3 + x + 1$
 (polynomial
 generator)

Divisor $x^3 + x + 1$ $\overline{) \begin{array}{r} x^3 + x \\ x^6 + x^3 \end{array}}$ ← Dividend
 $\underline{x^6 + x^4 + x^3}$
 x^4
 $\underline{x^4 + x^2 + x}$
 $x^2 + x$ → Remainder (degree is less than that of divisor)

Data unit to be transmitted	$x^6 + x^3$	$x^2 + x$
	Data	Remainder

CRC division using polynomial

Generator (divisor) polynomial: $g(x) = x^3 + x + 1$
 Information polynomial: $i(x) = x^3 + x^2$
 Dividend polynomial: $p(x) = x^3 i(x) = x^6 + x^5$

$$\begin{array}{r}
 x^3 + x + 1 \overline{) \begin{array}{r} x^3 + x^2 + x \\ x^6 + x^5 \\ x^6 + x^4 + x^3 \\ \hline x^5 + x^4 + x^3 \\ x^5 + x^3 + x^2 \\ \hline x^4 + x^2 \\ x^4 + x^2 + x \\ \hline x \end{array}} \\
 \end{array}$$

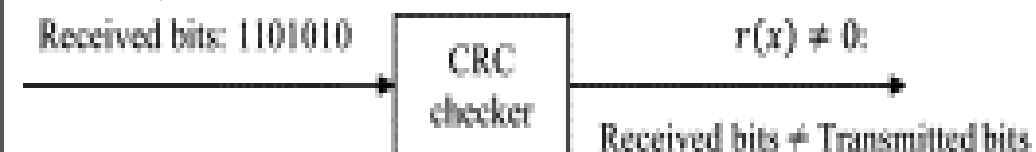
Remainder polynomial: $r(x) = x$
 Transmitted polynomial: $b(x) = p(x) + r(x) = x^6 + x^5 + x$



Generator (divisor) polynomial: $g(x) = x^3 + x + 1$
 Received (dividend) polynomial: $b'(x) = x^6 + x^5 + x^3 + x$

$$\begin{array}{r}
 x^3 + x + 1 \overline{) \begin{array}{r} x^3 + x^2 + x + 1 \\ x^6 + x^5 + x^3 + x \\ x^6 + x^4 + x^3 \\ \hline x^5 + x^4 + x \\ x^5 + x^3 + x^2 \\ \hline x^4 + x^3 + x^2 + x \\ x^4 + x^2 + x \\ \hline x^3 \\ x^3 + x + 1 \\ \hline x + 1 \end{array}} \\
 \end{array}$$

Remainder polynomial: $r(x) = x + 1$



Name	Polynomial	Application
CRC-8	$x^8 + x^2 + x + 1$	ATM header
CRC-10	$x^{10} + x^9 + x^5 + x^4 + x^2 + 1$	ATM AAL
CRC-16	$x^{16} + x^{12} + x^5 + 1$	HDLC
CRC-32	$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} +$ $x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$	LANs

R bits in CRC

In CRC the R bits are 1 bit less than the divisor

- Divisor 4 , $r = 3$
- Divisor 5 , $r = 4$

In Hamming code r is calculated using the formula

- **$2^r \geq k+r$**
- Case I: $8 > 7$ ---- $r=3$; $k=4$
- Case II: $k=5$, $r = 4$,
 $16 > 5+4$
- 8 possible combinations of error locations + no error = 9
- Syndrome bits options

For Cyclic Hamming Codes $2^r > k+r$

- k ? dataword r ?
- $k=2$, $r?$, $r=2$
- $2^2 > 2+2$ $2^3 > 2+3$ $r=3$ $8>5$
- $k=3$, $r?$ $r=3$, $8>6$
- $k=4$, $r?$ $r=3$, $8>7$
- $k=5$, $r?$ $r=3$, $8>5+3$, $r=4$, $16>8$
- $K=6$, $r=4$, $16>10$
- $K=7$, $r=4$, $16>11$